

Qty. Description

1 **MG160MA**



Note! Product picture may differ from actual product

Product No.: [87420033](#)

The motor is a 3-phase AC standard motor.

Grundfos motor technology represents the best from Grundfos within energy efficient motors. Grundfos motor either meet or exceed legislative requirements such as the EuP IE3 grade.

Technical:

Approvals for motor: CE,EAC,UKCA,SEPRO,RCMMorocco

Energy approvals for motor: CE

Measuring Equip: SVANTEK, SVAN 958

Cable gland entry: 2xM20 + 4xM40 knock out

Drain holes: Yes (closed)

Installation:

Range of ambient temperature: -30 .. 60 °C

Flange size for motor: SP.FM

Electrical data:

Motor standard: IEC

Motor type: 160MA

Rated power - P2: 11 kW

Mains frequency: 50 Hz

Rated voltage: 3 x 380-415D/660-690Y V

Rated current: 21,2-20,4/12,2-12,0 A

Maximum current consumption: 23,4-22,4/13,4-13,2 A

Starting current: 710-810 %

Cos phi - power factor: 0.86-0.81

Rated speed: 1470-1475 rpm

Rated full-load torque: 71.5 Nm

Locked-rotor torque: 180-210 %

Breakdown torque: 310-370 %

Moment of inertia: 0.058 kg m²

IE Efficiency class: IE3

Motor efficiency at full load: 91.4-91.4 %

Motor efficiency at 3/4 load: 92.1-91.9 %

Motor efficiency at 1/2 load: 91.6-91.4 %

Number of poles: 4

Enclosure class (IEC 34-5): 55 Dust/Jetting

Insulation class (IEC 85): F

Lubricant type: UNIREX N3

Lubricant quan.: 13 g

Relubrication interval 1: 14000 h / 40 °C

Relubrication interval 2: 3500 h / 60 °C

Bearing insulation type N-end: STEEL BEARING



Company name:

Created by:

Phone:

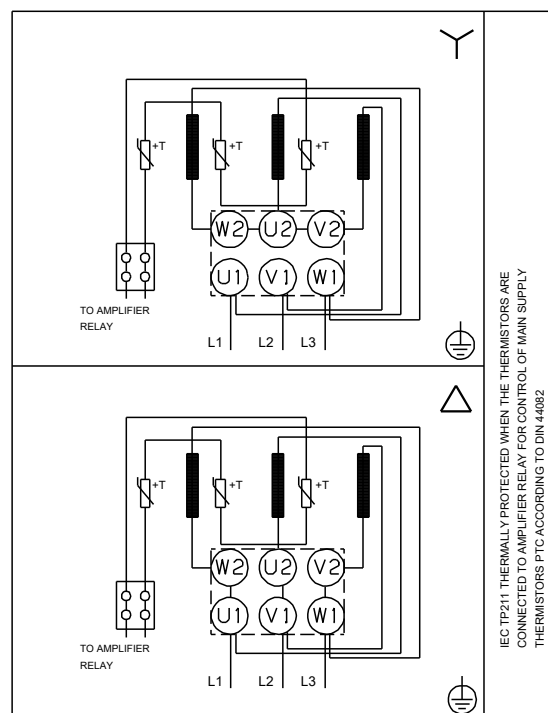
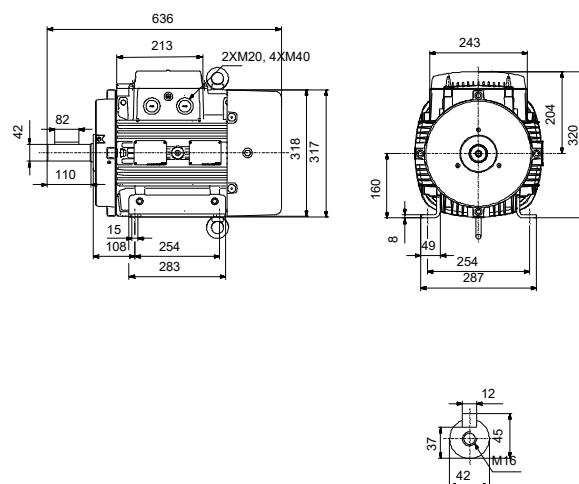
Date:

10/10/2023

Qty.	Description
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1	Cable gland entry: 2xM20 + 4xM40 knock out
	Others:
	Net weight: 106 kg
	Colour/Type: NCS 9000 gloss 40+-10 /E-coat
	Environmental approvals for motor: WEEE

Description	Value
General information:	
Product name:	MG160MA
Product No:	87420033
EAN number:	5710622822434
Technical:	
Shaft diameter:	42 mm
Length of shaft end:	110 mm
Approvals for motor:	CE,EAC,UKCA,SEPRO,RCMMor occo
Energy approvals for motor:	CE
Model:	H3
Cooling:	IC 411
Measuring Equip:	SVANTEK, SVAN 958
Cable gland entry:	2xM20 + 4xM40 knock out
Drain holes:	Yes (closed)
Installation:	
Range of ambient temperature:	-30 .. 60 °C
Flange size for motor:	SP.FM
Electrical data:	
Motor standard:	IEC
Motor type:	160MA
Rated power - P2:	11 kW
Mains frequency:	50 Hz
Rated voltage:	3 x 380-415D/660-690Y V
Rated current:	21,2-20,4/12,2-12,0 A
Maximum current consumption:	23,4-22,4/13,4-13,2 A
Starting current:	710-810 %
Cos phi - power factor:	0.86-0.81
Rated speed:	1470-1475 rpm
Rated full-load torque:	71.5 Nm
Locked-rotor torque:	180-210 %
Breakdown torque:	310-370 %
Moment of inertia:	0.058 kg m ²
IE Efficiency class:	IE3
Motor efficiency at full load:	91.4-91.4 %
Motor efficiency at 3/4 load:	92.1-91.9 %
Motor efficiency at 1/2 load:	91.6-91.4 %
Number of poles:	4
Enclosure class (IEC 34-5):	55 Dust/Jetting
Insulation class (IEC 85):	F
Built-in motor protection:	PTC
Tropic protec:	no
Thermal protec:	external
R1:	0.72 ohm
R2:	0.72 ohm
R3:	0.72 ohm
Winding 1:	PHASE 1
Winding 2:	PHASE 2
Winding 3:	PHASE 3
Mount. design. acc. IEC 34-7:	IM B3
Lubricant type:	UNIREX N3
Lubricant quan.:	13 g
Temperature 1:	14000 h / 40 °C
Temperature 2:	3500 h / 60 °C
Bearing insulation type N-end:	STEEL BEARING
Cable gland entry:	2xM20 + 4xM40 knock out
Others:	
Net weight:	106 kg



IEC TP211 THERMALLY PROTECTED WHEN THE THERMISTORS ARE
CONNECTED TO AMPLIFIER RELAY FOR CONTROL OF MAIN SUPPLY
THERMISTORS PTC ACCORDING TO DIN 4482



Company name:

Created by:

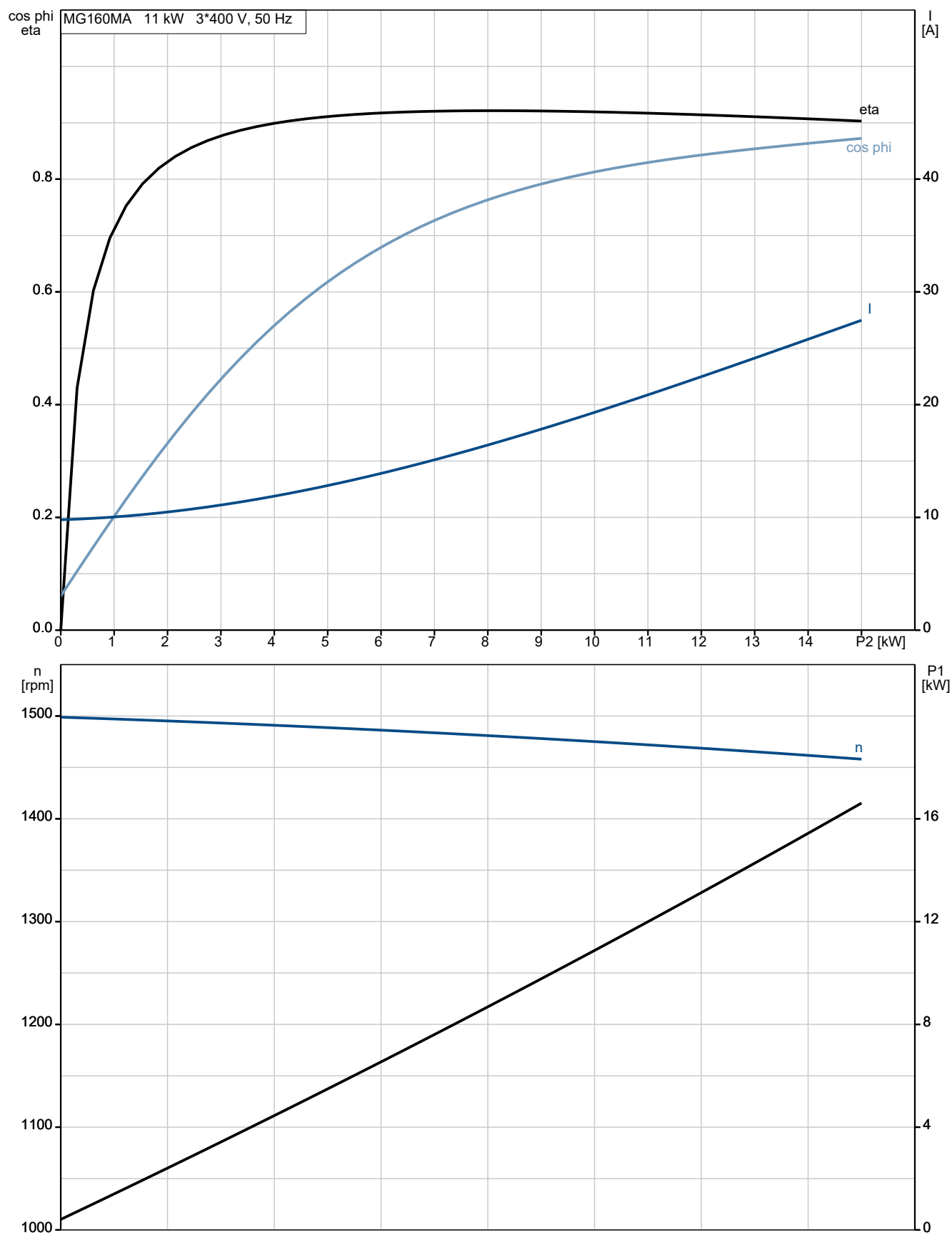
Phone:

Date:

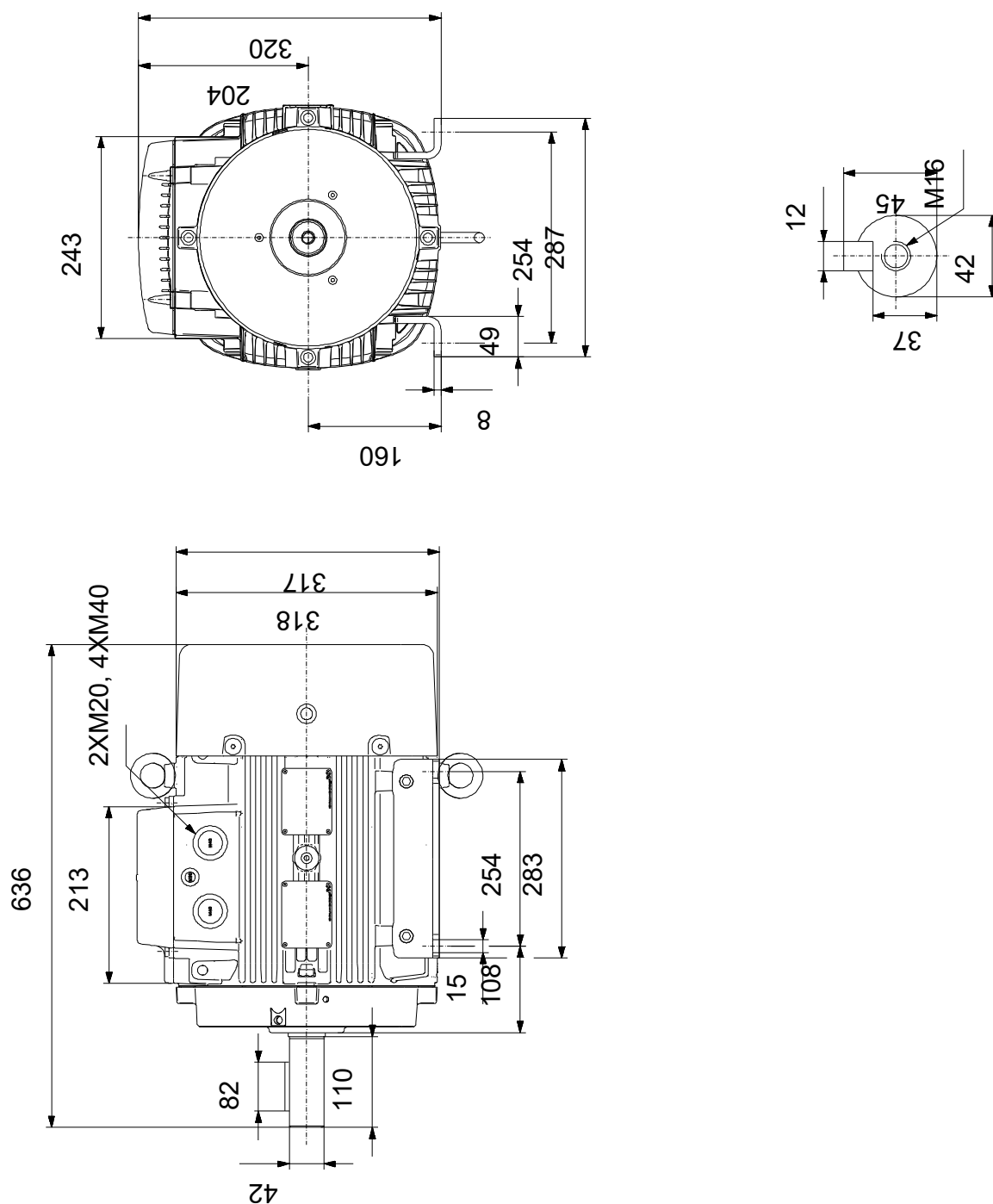
10/10/2023

Description	Value
Colour/Type:	NCS 9000 gloss 40+-10 /E-coat
Environmental approvals for motor:	WEEE

87420033 MG160MA 50 Hz



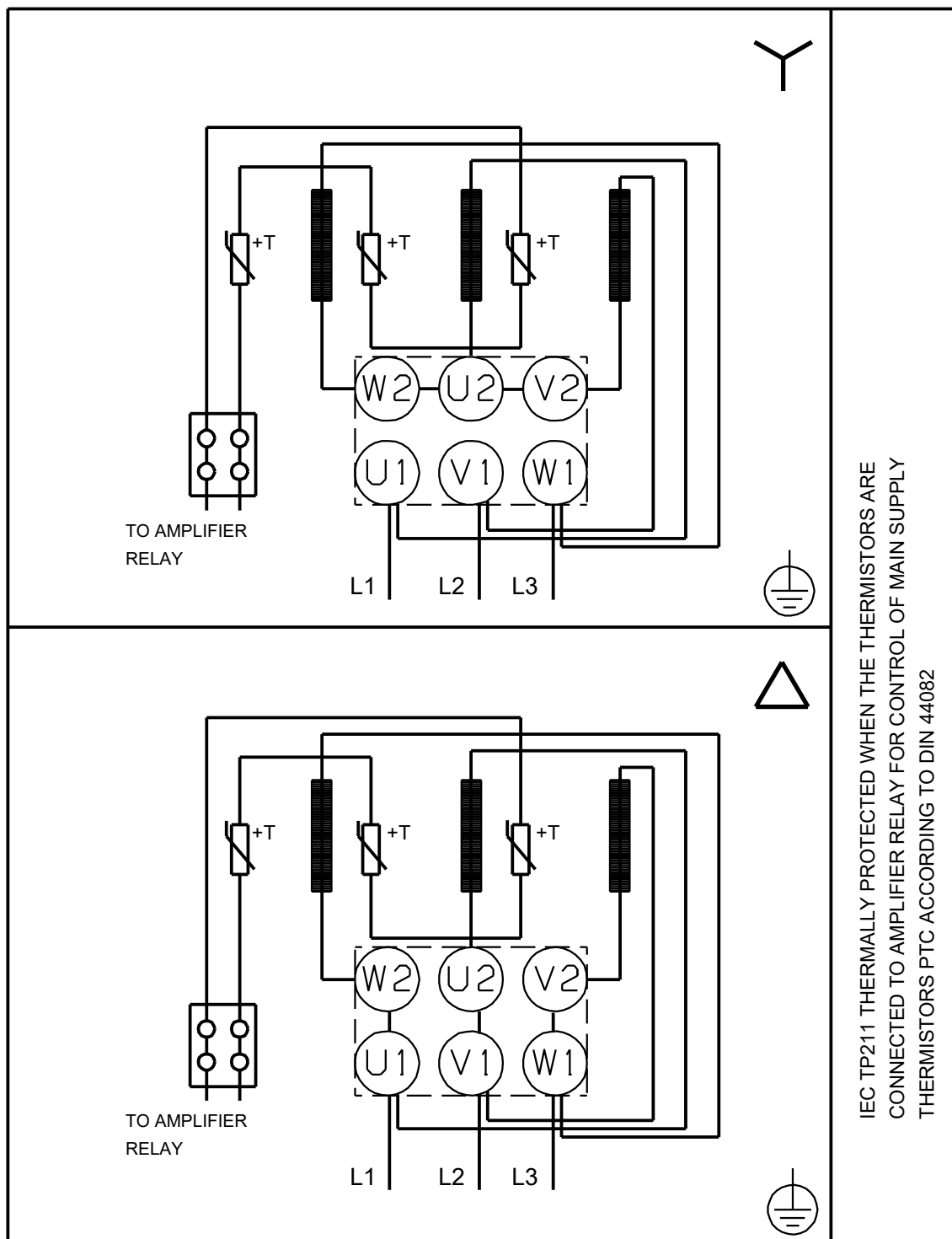
87420033 MG160MA 50 Hz



Note! All units are in [mm] unless others are stated.

Disclaimer: This simplified dimensional drawing does not show all details.

87420033 MG160MA 50 Hz



IEC TP211 THERMALLY PROTECTED WHEN THE THERMISTORS ARE
CONNECTED TO AMPLIFIER RELAY FOR CONTROL OF MAIN SUPPLY
THERMISTORS PTC ACCORDING TO DIN 44082

Note! All units are in [mm] unless others are stated.

Ordering data

MG160MA4-42-H3

Company name:

Created by:

Phone:

Date:

Fax:

Product No.: **87420033**

10/10/2023

Project:

Contact:

Client Number:

Reference Number:



Electrical data:

Frequency	50 Hz
Rated motor voltage	380-415D/660-690Y V
Rated motor power	11 kW
Rated motor speed	1470-1475 rpm
Rated motor torque	71.5 Nm
Wiring connection	D/Y
Rated motor current	21,2-20,4/12,2-12,0 A
Max motor current	23,4-22,4/13,4-13,2 A
Efficiency class	IE3
Full load Efficiency %	91.4-91.4 %
Power factor	0,86-0,81
Starting- / Rated motor current (50/60 Hz)	710 - 810 /
Breakdown- / Rated motor torque (50/60 Hz)	310-370 / %
Starting- / Rated motor torque (50/60 Hz)	180 - 210 /

Mechanical data:

Noise measurements (50/60 Hz)	58,0 dB(A)
Moment of inertia	0.058 kg m ²
Drain holes	Yes (open)
Bearing DE	6309.Z.C4
Bearing NDE	6309.Z.C4
Bearing arrangement	DE-locked
Type of bearing DE	Steel ball
Bearing seal DE	-
Characteristic of grease DE	-
Type of bearing NDE	Steel ball
Bearing seal NDE	-
Characteristic of grease NDE	-
Regreasing device	Yes
Type of lubrication	UNIREX N3
Relubrication interval	14000
Quantity of grease for relubrication	13 g
External earthing	NO
Colour/type	NCS 9000 gloss 40+-10 /E-coat

Explosion protection:

Type of protection WITHOUT (STANDARD)

Ambient conditions:

Ambient temperature -30°C to 60°C

Altitude above sea level 1000 m

Approvals and specifications IEC 60034

General data:

Frame size	160MA
Type of construction	IM B3 Flange size SP.FM
Weight, without optional accessories	106 kg
Frame material	Aluminium
Degree of protection	IP 55 Dust/Jetting
Method of cooling, TEFC	IC 411
Vibration class	A (Standard)
Insulation class	155(F) to 130(B)
Duty type	S1 = Continuous operation
Direction of rotation	CCW / UNI-DIRECTIONAL

Terminal box:

Material of terminal box	-
Cable entry	-
Cable gland	-

Protection:

Built-in protection Protector type PTC 160 degC

87420033 MG160MA 50 Hz

Vibration Measurement Report

Mechanical vibration measurement according to IEC 60034-14

Motor type: MG160MA

Product number: 87420033

Fan diameter: D215 [mm]

P2: 11 [kW]

U: 400 [V]

Pole: 4

Frequency: 50 [Hz]

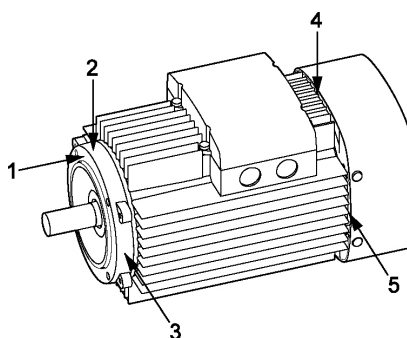
Remark: According to IEC 60034-14 the motor is tested at no load with all supply quantities at their rated value

Measuring equipment: SVANTEK, SVAN 958

Measuring method:

Mounting: Free suspension

Measuring point	[mm/s] RMS
1	0,20
2	0,50
3	0,50
4	0,45
5	0,60



Measuring points as indicated on the drawing

Limits of vibration severity

Limits of maximum vibratio velocity from IEC 60034-14				
Vibration grade	Mounting	Frame size H in [mm]		
		56 <= H <= 132	132 < H <= 280	280 < H
A	Free suspension	1,6	2,2	2,8
	Rigid mounting	1,3	1,8	2,3
B	Free suspension	0,7	1,1	1,8
	Rigid mounting	--	0,9	1,5

Grade 'A' applies to machines with no special vibration requirements

Grade 'B' applies to machines with special vibration requirements

Initials: BAO

File no.: 04.03.2013

87420033 MG160MA 50 Hz

Sound Measurement Report

Sound Power measurement according to DS/ISO-3743

Motor type: MG160MA

Product number: 87420033

Fan diameter: D215 [mm]

P2: 11 [kW]

U: 400 [V]

Pole: 4

Frequency: 50 [Hz]

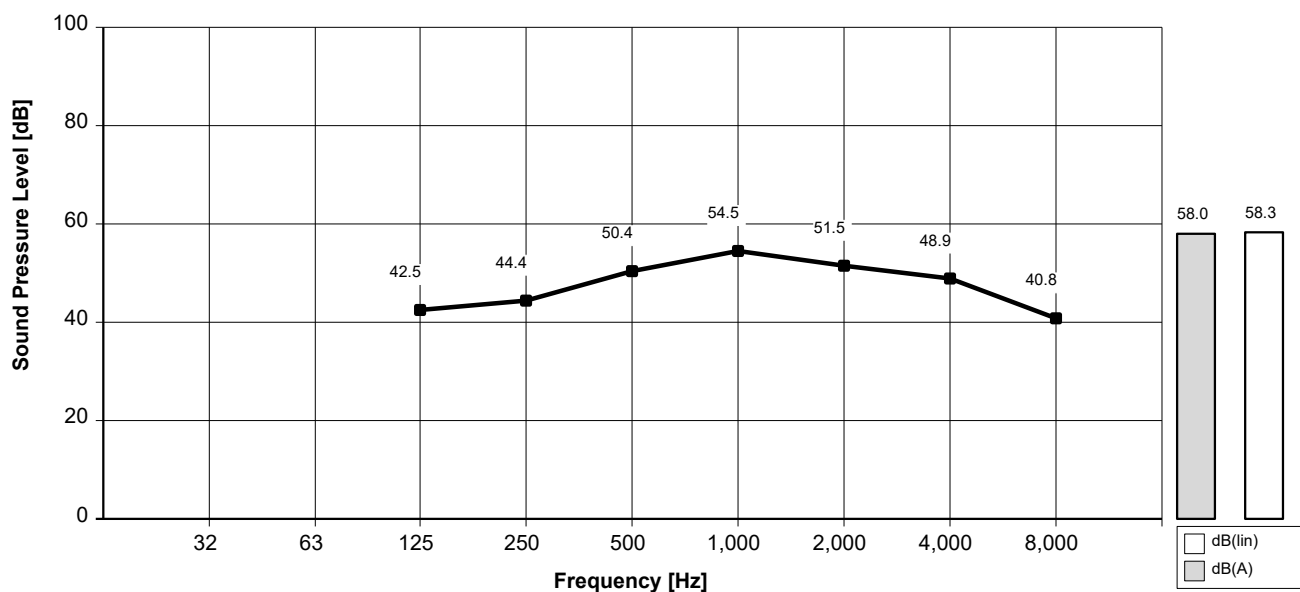
Remarks: The declared noise emission value, does not take production variations and measurement uncertainties into account. The declared value may therefore be up-to 3 dB higher, than the value for the average production unit.

Tested at:

Octave Band Level [db]

Center Frequency	125	250	500	1000	2000	4000	8000	dB(A)	dB(lin)
Sound Power Level re: 1 pW	55,3	57,3	63,2	67,3	64,4	61,7	53,7	70,8	71,1
Sound Pressure Average at 1m. re: 20 UPa	42,5	44,4	50,4	54,5	51,5	48,9	40,8	58,0	58,3

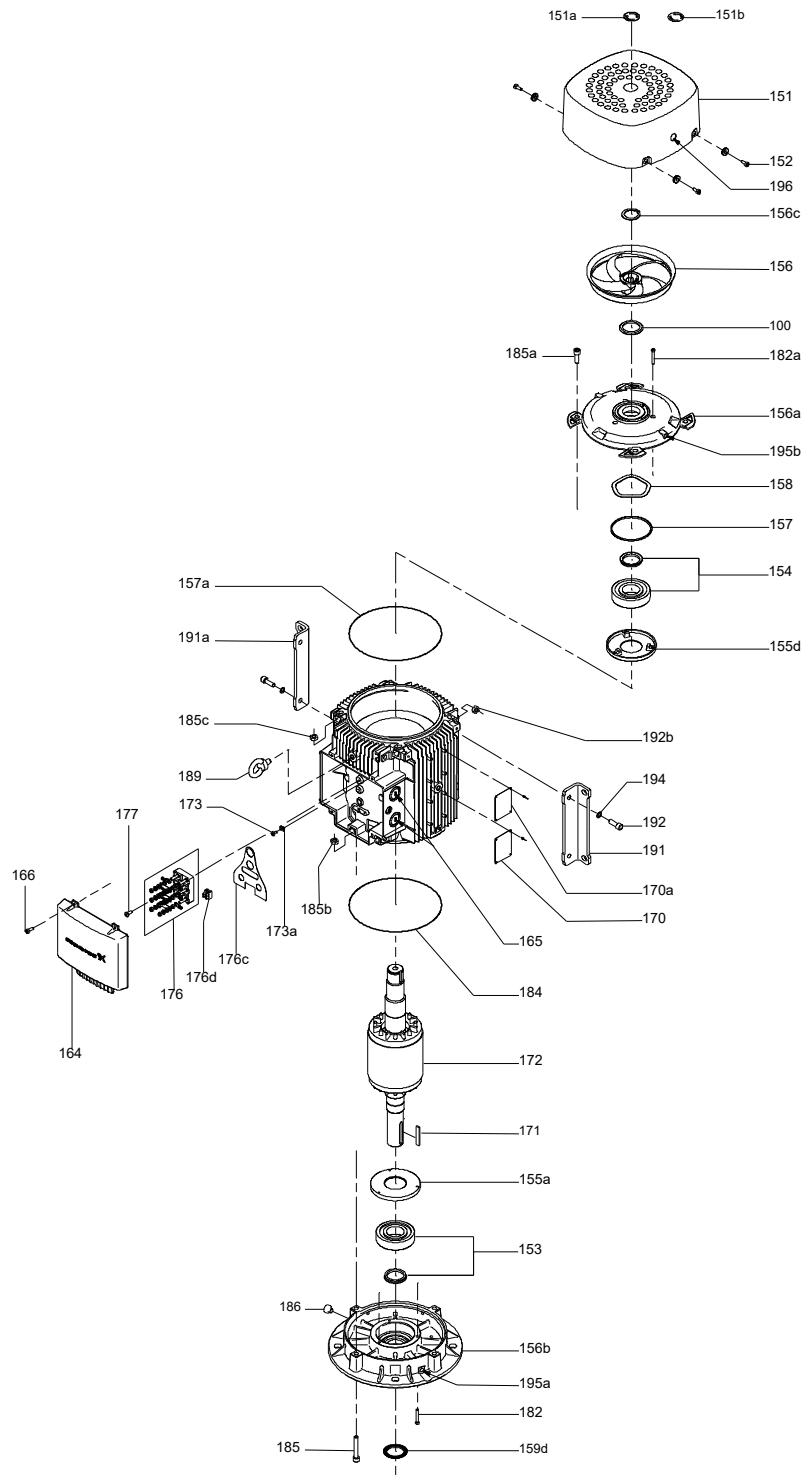
Sound Pressure calculated according to ISO/DIS 11203 method Q2



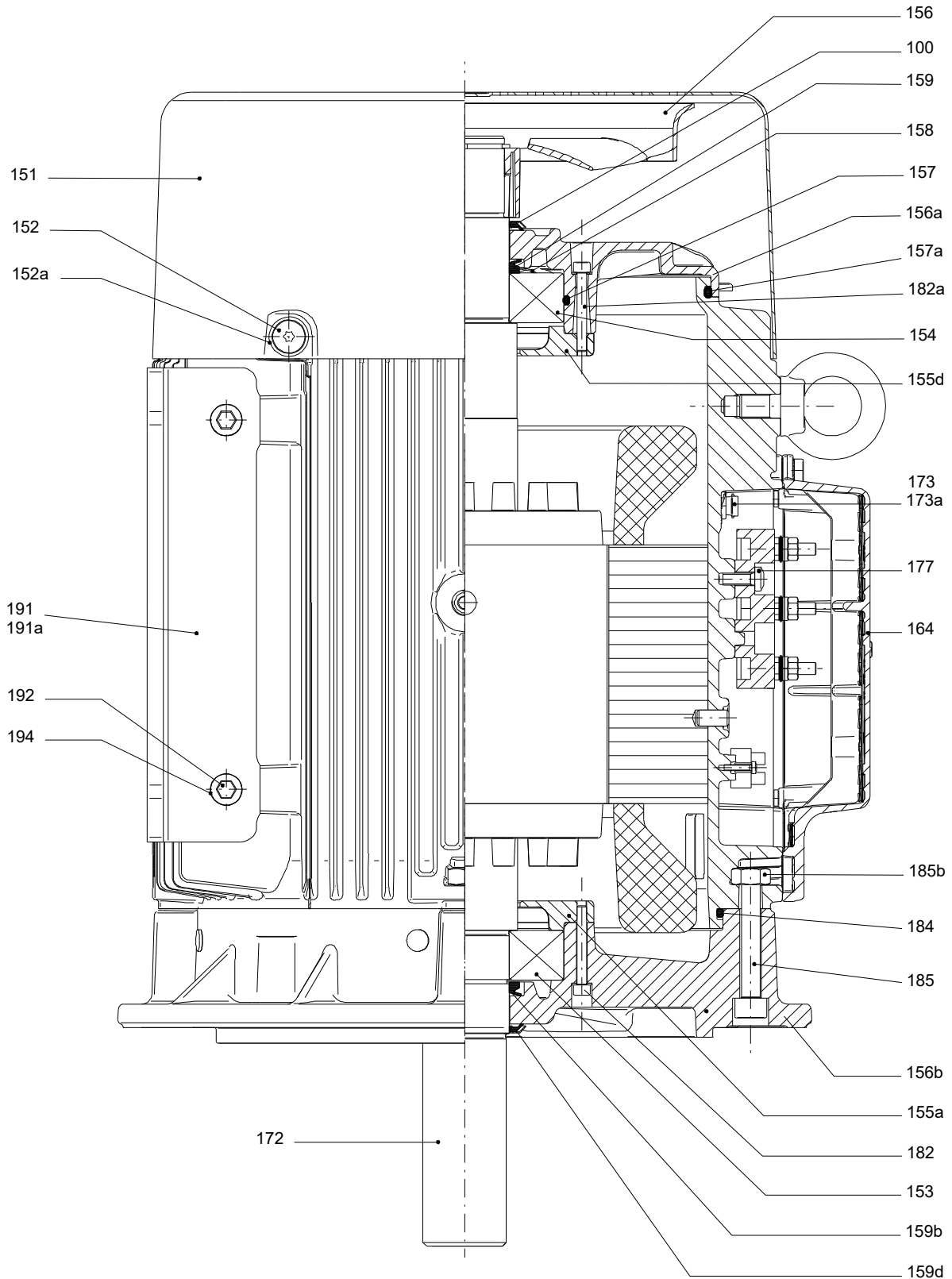
Initials: KIR

File no.: 02.05.2013

Exploded view



Sectional drawing



Spare parts MG160MA, Product No. 87420033
Produced After 1152 (production year and week number)

Pos	Description	Annotation	Classification Data	Part no.	Qty.	Unit
-	Kit, Bearing cpl.			96796674	1	pcs
32b	Waved washer				1	pcs
153	Ball bearing				1	pcs
154	Ball bearing				1	pcs
157	O-ring				1	pcs
-	Kit, Bearing plate			96796664	1	pcs
155.a	Bearing cover				1	pcs
208a	Gasket				3	pcs
208	Hex socket head cap screw		Designation: DIN912		3	pcs
			Length (mm): 40			
			Thread: M5			
-	Kit, End shield			96796660	1	pcs
156.b	End shield DE				1	pcs
159.b	Seal ring				1	pcs
185b	Nut				4	pcs
185	Hex socket head cap screw				4	pcs
186	Drain plug				3	pcs
195a	Grease nipple				1	pcs
-	Kit, Eyebolt			96796712	1	pcs
189	Eyebolt				2	pcs
-	Kit, Fan			96796652	1	pcs
156.c	Retaining ring				1	pcs
156	Fan				1	pcs
-	Kit, fan cover, black			96796647	1	pcs
151	Fan cover				1	pcs
152	Pan head thread forming screw				4	pcs
196	Nipple				1	pcs
-	Kit, fan cover, black/red			92891507	1	pcs
151	Fan cover				1	pcs
152	Pan head thread forming screw				4	pcs
-	Kit, Foot			96796666	1	pcs
185b	Nut				4	pcs
191a	Foot				1	pcs
191	Foot				1	pcs
192	Hex socket head cap screw				4	pcs
194	Lock washer				4	pcs
-	Kit, Gaskets			96798508	1	pcs
184	O-ring		Diameter: 235		2	pcs
-	Kit, Lubrication nipple			96796671	1	pcs
195b	Grease nipple				2	pcs
195a	Grease nipple				1	pcs
-	Kit, ND-end shield cpl.			96796669	1	pcs
32b	Waved washer				1	pcs
156.a	End shield NDE				1	pcs
157	O-ring				1	pcs
159	V-ring				1	pcs
185.c	Nut				4	pcs
185.a	Hex socket head cap screw				4	pcs
195b	Grease nipple				1	pcs
-	Kit, Shaft seal			96843459	1	pcs
159.b	V-ring				1	pcs
159	V-ring				1	pcs
-	Kit, Terminal board			96796657	1	pcs



Company name:

Created by:

Phone:

Date: 10/10/2023

Pos	Description	Annotation	Classification Data	Part no.	Qty.	Unit
	Terminal connection				3	pcs
	Washer				12	pcs
36	Hex nut				12	pcs
173	Torx Screw				1	pcs
176.d	Terminal board				1	pcs
176	Terminal board				1	pcs
177	Torx Screw				1	pcs
-	Kit, Terminal box cover			96796659	1	pcs
164	Terminal box cover w/gasket				1	pcs
166	Pan head thread forming screw				4	pcs
-	Spare, Key			96279873	1	pcs
40	Parallel key		Designation: DS96 DIN 6885 Dimension: 12X8X90		1	pcs